

Quantum Entanglement & Quantum Computing Units 3 & 4 Question Bank

Photonics and Quantum Technology Lab
ALL THE BEST

UNIT 3: ENTANGLEMENT AND QUANTUM INFORMATION

1-Mark Multiple Choice Questions (16 Questions)

Question 1.1.1 Which of the following states is NOT a Bell state?

A $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

B $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

C $\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle)$

D $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

Answer: C

Solution (1 mark): Bell states are maximally entangled two-qubit states. Options A, B, D are the four Bell basis states. Option C is a product-like state (separable up to normalization), not entangled.

Question 1.1.2 In quantum entanglement, when two qubits are maximally entangled, measuring one qubit:

A Instantly determines other (violates no-signaling)

B Gives complete info about both

C Provides correlations but NO FTL communication

D Random result independent of other

Answer: C

Solution (1 mark): Entanglement creates perfect correlations, but measurement outcome is random locally. Cannot be used for faster-than-light signaling—no information transmission occurs.

Question 1.1.3 Schmidt decomposition $|\psi\rangle_{AB} = \sum_i \lambda_i |\psi\rangle_A^i |\psi\rangle_B^i$ with $\sum_i \lambda_i^2 = 1$. What is λ_i ?

- A Schmidt coefficients (probabilities)
- B Eigenvalues of reduced density matrices
- C Square roots of probabilities
- D All of above

Answer: D

Solution (1 mark): λ_i are eigenvalues of $\rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|)$, represent Schmidt coefficients, and form probability distribution $\sum_i \lambda_i^2 = 1$.

Question 1.1.4 SPDC generates entangled photons by:

- A Splitting single photon into two (energy-momentum conserving)
- B Entangling pre-existing photons
- C Filtering from thermal source
- D Stimulated emission

Answer: A

Solution (1 mark): SPDC in nonlinear crystals: pump photon splits into signal and idler photons. Conserves energy $\omega_p = \omega_s + \omega_i$ and momentum $\vec{k}_p = \vec{k}_s + \vec{k}_i$.

Question 1.1.5 Most practical entanglement type for quantum communication systems:

- A Time-energy
- B Orbital angular momentum (OAM)
- C Polarization
- D Frequency

Answer: C

Solution (1 mark): Polarization entanglement is easy to generate via SPDC, manipulate with standard optics (polarizers, waveplates), and robust in free-space links. Standard for QKD systems (BB84, E91).

Question 1.1.6 Purity $P = \text{Tr}(\rho^2)$ for maximally mixed state $\rho = \frac{I_2}{2}$ (2-qubit):

- A $P = 0$
- B $P = 0.5$
- C $P = 1$
- D Undefined

Answer: B

Solution (1 mark): $\text{Tr}((\frac{I_2}{2})^2) = \text{Tr}(\frac{I_4}{4}) = \frac{4}{4} = 1 \rightarrow \frac{1}{4}$. Pure state: $P = 1$; maximally mixed d -dim: $P = \frac{1}{d}$.

Question 1.1.7 For maximally entangled two-qubit state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, reduced density of one qubit:

- A Pure $|0\rangle$
- B Maximally mixed $\frac{I_2}{2}$
- C Pure superposition
- D Depolarized

Answer: B

Solution (1 mark): Trace out qubit 2: $\rho_1 = \text{Tr}_2(|\Phi^+\rangle\langle\Phi^+|) = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I_2}{2}$ (maximally mixed). Purity $\text{Tr}(\rho_1^2) = \frac{1}{2} < 1$.

Question 1.1.8 Shannon entropy $H(X) = -\sum_x p(x) \log_2 p(x)$ maximized when:

- A $p(x) = \frac{1}{N}$ (uniform)
- B One outcome $p_1 = 1$, others 0
- C Exponential decay
- D Linear increase

Answer: A

Solution (1 mark): Entropy is maximum for uniform distribution by convexity of $-\sum_x x \log x$. $H_{\max} = \log_2 N$ bits (all outcomes equally likely).

Question 1.1.9 Von Neumann entropy $S(\rho) = -\text{Tr}(\rho \log_2 \rho)$ for pure state:

- A Maximum = $\log_2 d$
- B Zero
- C One
- D State-dependent

Answer: B

Solution (1 mark): Pure state has eigenvalues $\{1, 0, 0, \dots\}$. So $S = -1 \cdot \log_2 1 - 0 \cdot \log_2 0 = 0$.

Question 1.1.10 No-cloning theorem states:

- A Any state clones with 50% success
- B No universal machine clones arbitrary unknown states
- C Known states uncloneable
- D Cloning only for entangled

Answer: B

Solution (1 mark): Fundamental result: no unitary U exists such that $U|\psi\rangle|\text{blank}\rangle = |\psi\rangle|\psi\rangle$ for all unknown $|\psi\rangle$. Known states CAN be cloned deterministically.

Question 1.1.11 Quantum dense coding: Alice sends classical bits by transmitting:

- A 1 bit, 1 qubit
- B 2 bits, 1 qubit
- C 4 bits, 2 qubits
- D 1 bit, 2 qubits

Answer: B

Solution (1 mark): Protocol: Alice-Bob share Bell pair. Alice applies one of 4 unitaries (I, X, Z, XZ) to her qubit and sends it to Bob. Bob Bell-measures and extracts 2 classical bits from 1 transmitted qubit.

Question 1.1.12 Quantum teleportation requires:

- A Pre-shared Bell pair only
- B 2 classical bits only
- C Bell pair + 2 classical bits
- D Pre-existing entanglement only

Answer: C

Solution (1 mark): Alice measures her qubit and half of Bell pair, yielding 2 classical bits. Sends bits to Bob. Bob applies correction unitary. Requires BOTH resources—no FTL possible.

Question 1.1.13 Born rule: probability of measuring eigenvalue a is:

- A $p_a = |\langle a|\psi\rangle|^2$
- B $p_a = \langle a|\psi\rangle$
- C $p_a = |\psi\rangle^2$
- D $p_a = \frac{1}{a}$

Answer: A

Solution (1 mark): Born rule: If $A|a\rangle = a|a\rangle$ and $|\psi\rangle = \sum_i c_i |a_i\rangle$, then $p_a = |c_a|^2 = |\langle a|\psi\rangle|^2$.

Question 1.1.14 Maximal two-qubit entanglement (Bell state) corresponds to single-qubit entropy:

- A $S_A = 0$
- B $S_A = 1$ bit
- C $S_A = 2$ bits
- D Basis-dependent

Answer: B

Solution (1 mark): Bell state $|\Phi^+\rangle$: $\rho_A = \frac{I_2}{2}$ (maximally mixed single qubit). $S_A = -\text{Tr}(\rho_A \log_2 \rho_A) = 1$ bit.

Question 1.1.15 CHSH inequality violation $S \leq 2$ (max classical) vs $S \leq 2\sqrt{2}$ (quantum) demonstrates:

- A Quantum correlations stronger than classical
- B Non-locality
- C Bell inequality violation
- D All above

Answer: D

Solution (1 mark): CHSH $S = 2$ classically, quantum maximum $S = 2\sqrt{2} \approx 2.828$ for entangled state. Proves no local hidden variable model.

Question 1.1.16 Quantum error correction code protecting against single bit-flip:

- A 3-qubit Repetition code
- B Shor code (9 qubits)
- C Surface code
- D Bacon-Shor

Answer: A

Solution (1 mark): Repetition code: $|0\rangle \rightarrow |000\rangle$, $|1\rangle \rightarrow |111\rangle$. Detects/corrects any single-qubit bit-flip. Minimum distance $d = 3$ corrects $\frac{d-1}{2} = 1$ error.

2-Mark Multiple Choice Questions (12 Questions for Unit 3)

Question 1.2.1 Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Measure qubit 1 as 0. Qubit 2 becomes:

- A 0 (certainly)
- B 1 (certainly)
- C Maximally mixed
- D Undetermined

Answer: B

SOLUTION (2 marks): When qubit 1 measured as 0, state collapses to $|01\rangle$ (only basis state with first qubit 0). Therefore qubit 2 is definitely in state 1.

Mark breakdown:

- 0.5 mark: Collapse concept
- 0.5 mark: Identifying basis state
- 1 mark: Final answer (certainty)

Question 1.2.2 System evolves $U = e^{-iHt/\hbar}$ with $H = \sigma_z$, $\hbar = 1$. Start $|\psi\rangle = |0\rangle$, time $t = \frac{\pi}{2}$. Result:

- A $|0\rangle$
- B $|1\rangle$
- C $e^{-i\pi/4}|0\rangle$
- D $-i|0\rangle$

Answer: C

SOLUTION (2 marks): $U = e^{-i\sigma_z\pi/4}$. Since $\sigma_z|0\rangle = |0\rangle$ (eigenstate with eigenvalue 1): $e^{-i\sigma_z\pi/4}|0\rangle = e^{-i(\pi/4)}|0\rangle = e^{-i\pi/4}|0\rangle$

Mark breakdown:

- 0.5 mark: Eigenstate recognition
- 1 mark: Phase calculation
- 0.5 mark: Simplification

Question 1.2.3 Mutual information $I(A : B) = S_A + S_B - S_{AB}$ for Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$:

- A 0 bits
- B 1 bit
- C 2 bits
- D Undefined for quantum

Answer: C

SOLUTION (2 marks): Formula $I(A : B) = S_A + S_B - S_{AB}$. For Bell state: $S_A = 1$ bit (reduced $\rho_A = \frac{I_2}{2}$), $S_B = 1$ bit (reduced $\rho_B = \frac{I_2}{2}$), $S_{AB} = 0$ (pure state). Thus $I(A : B) = 1 + 1 - 0 = 2$ bits.

Mark breakdown:

- 0.5 mark: Formula
- 0.75 mark: Individual entropies
- 0.75 mark: Final calculation

Question 1.2.4 CNOT on state $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ (control first):

- A Separable product
- B $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ (Bell)
- C Mixed state
- D Maximally mixed

Answer: B

SOLUTION (2 marks): Initial state: $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$. CNOT logic: if control qubit is 1, flip target; else no change.

$$|00\rangle \xrightarrow{\text{CNOT}} |00\rangle \text{ (no flip)}$$

$$|10\rangle \xrightarrow{\text{CNOT}} |11\rangle \text{ (flip target)}$$

Result: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle$ (maximally entangled Bell state).

Mark breakdown:

- 0.5 mark: CNOT mechanics
- 0.5 mark: Gate application
- 1 mark: Final entangled state identification

Question 1.2.5 Two-qubit entangled state $|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |11\rangle)$. Entanglement of formation approximately:

- A 0 bits
- B 0.5 bits
- C 1 bit
- D Cannot be determined

Answer: B

SOLUTION (2 marks): Reduced density matrix $\rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|) = \begin{pmatrix} 2/3 & 0 \\ 0 & 1/3 \end{pmatrix}$. Entropy $S_A = -\frac{2}{3}\log_2(\frac{2}{3}) - \frac{1}{3}\log_2(\frac{1}{3}) \approx 0.918$ bits. Entanglement of formation estimates to roughly 0.5 bits for this state.

Mark breakdown:

- 0.5 mark: Reduced density matrix
- 1 mark: Entropy calculation
- 0.5 mark: Entanglement estimate

Question 1.2.6 W-state $|W\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$ bipartite entanglement (Qubit 1 vs. Qubits 2,3):

- A Maximally entangled
- B Separable
- C Partially entangled
- D No entanglement

Answer: C

SOLUTION (2 marks): Reduced $\rho_1 = \text{Tr}_{2,3}(|W\rangle\langle W|) = \frac{2}{3}|0\rangle\langle 0| + \frac{1}{3}|1\rangle\langle 1|$. This is mixed (not separable into product). Entropy $S_1 = -\frac{2}{3}\log_2(\frac{2}{3}) - \frac{1}{3}\log_2(\frac{1}{3}) \approx 0.918$ bits (partial).

Mark breakdown:

- 0.75 mark: Reduced density calculation
- 0.75 mark: Mixedness identification
- 0.5 mark: Partial entanglement conclusion

Question 1.2.7 Concurrence $C(\rho)$ for two-qubit state $\rho = \frac{1}{2}|\Phi^+\rangle\langle\Phi^+| + \frac{1}{2}|00\rangle\langle 00|$:

- A 0 (separable)
- B 0.5
- C 1 (maximally entangled)
- D Cannot be determined

Answer: B

SOLUTION (2 marks): Mixture: 50% Bell state (fully entangled, $C = 1$) + 50% product state (separable, $C = 0$). Average concurrence $C(\rho) \approx 0.5$.

Mark breakdown:

- 1 mark: Identifying mixture components
- 0.75 mark: Concurrence of each component
- 0.25 mark: Averaging for mixed state

Question 1.2.8 GHZ state $|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ vs. Bell pair triplication, multipartite entanglement:

- A Tripartite entanglement (GHZ) stronger
- B Three Bell pairs have more total entanglement
- C Equivalent entanglement
- D Cannot compare

Answer: B

SOLUTION (2 marks): GHZ: monogamy of entanglement restricts pairwise sharing. Three Bell pairs: $C_{total} = 3$ (sum of three pairs). GHZ: true 3-body correlations but limited bipartite entanglement. Three Bell pairs give more total two-qubit entanglement.

Mark breakdown:

- 0.75 mark: GHZ multipartite structure
- 0.75 mark: Three-Bell-pair analysis
- 0.5 mark: Comparison and conclusion

Question 1.2.9 Quantum steering: Alice measures and announces result; Bob's state changes without direct measurement on his system. Einstein called this:

- A "Spooky action at a distance"
- B Non-separability
- C Entanglement
- D All above

Answer: A

SOLUTION (2 marks): Steering (Schrödinger's term) involves asymmetric non-locality: one party (Alice) steers distant system's state without Bob's direct measurement. Related to Einstein's EPR paradox language about "spooky action."

Mark breakdown:

- 1 mark: Steering concept
- 0.75 mark: Einstein historical context
- 0.25 mark: Answer selection

Question 1.2.10 For maximally entangled state $|\Phi^+\rangle$, if Alice measures her qubit in basis $\{|+\rangle, |-\rangle\}$ (Hadamard), probability of measuring $|+\rangle$:

- A 0
- B 0.5
- C 1
- D $\frac{1}{\sqrt{2}}$

Answer: B

SOLUTION (2 marks): $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|+\rangle|+\rangle + |-\rangle|-\rangle)$ (in Hadamard basis). Probability of each outcome: 0.5.

Mark breakdown:

- 1 mark: Basis transformation
- 0.5 mark: Probability calculation
- 0.5 mark: Correct answer

Question 1.2.11 Continuous variable (CV) entanglement: two-mode squeezed state. Entanglement measure:

- A Logarithmic negativity
- B Concurrence
- C EPR variance
- D All apply

Answer: D

SOLUTION (2 marks): CV entanglement quantified by: logarithmic negativity (eigenvalues of partially transposed covariance), EPR variance (correlations), and for specific cases concurrence analogs. All three measures apply to different aspects of CV entanglement.

Mark breakdown:

- 0.75 mark: Recognizing CV framework
- 0.75 mark: Identifying applicable measures
- 0.5 mark: Answer selection

Question 1.2.12 Entanglement swapping protocol: Alice-Bob pair, Charlie-Diana pair (both Bell). After Charlie-Alice Bell measurement, Alice-Diana become entangled. This demonstrates:

- A Transitivity of entanglement
- B Monogamy violation
- C Entanglement is not transitive
- D Genuine monogamy requirement

Answer: A

SOLUTION (2 marks): Swapping shows entanglement "transitivity": $E(A, B) + E(B, C) \rightarrow E(A, C)$ after Bell measurement. Monogamy is preserved: $E(A : BC) + E(B : AC) \leq \text{constant}$. This is a key resource for quantum networks.

Mark breakdown:

- 1 mark: Protocol explanation
- 0.75 mark: Transitivity concept
- 0.25 mark: Answer selection

4-Mark Subjective Questions ($5 \times 4\text{M} = 20$ marks for Unit 3)

[NUMERICAL QUESTIONS: 3 per block]

Question 1.3.1 – NUMERICAL (4 marks)

Entanglement Characterization

Two-qubit system:

$$|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

- (a) Compute reduced density matrix ρ_A . (1.5 marks)
- (b) Calculate entropy $S_A = -\text{Tr}(\rho_A \log_2 \rho_A)$. (1 mark)
- (c) Determine if state is entangled and justify. (1.5 marks)

(a) Reduced density matrix (1.5 marks)

Full density matrix:

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Tracing out qubit B:

$$\rho_A = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

(b) Entropy (1 mark)

Eigenvalues of ρ_A :

$$\lambda = \{1, 0\}$$

Thus

$$S_A = -1 \log_2 1 = 0.$$

(c) Entanglement test (1.5 marks)

Since $S_A = 0$, subsystem A is in a pure state $\rightarrow$ **no entanglement**. Moreover:

$$|\psi\rangle = |+\rangle \otimes |+\rangle$$

with $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \rightarrow$ **product state, Schmidt rank = 1.**

Therefore: **State is separable, not entangled.**

Question 1.3.2 – NUMERICAL (4 marks)**Multipartite Entanglement: GHZ State**

Consider:

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

- (a) Compute $\rho_A = \text{Tr}_{BC}(|GHZ\rangle\langle GHZ|)$. (1.5 marks)
- (b) Find von Neumann entropy S_A . (1 mark)
- (c) Comment on GHZ vs Bell-pair entanglement. (1.5 marks)

(a) Reduced density (1.5 marks)

Non-zero contributions from tracing out B,C:

$$\rho_A = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I_2}{2}$$

(b) Entropy (1 mark)

$$S_A = -\text{Tr} \left(\frac{I_2}{2} \log_2 \frac{I_2}{2} \right) = 1 \text{ bit}$$

(c) Interpretation (1.5 marks)

- Each single qubit is maximally mixed $\rightarrow$ strong multipartite entanglement.
- GHZ correlations vanish if one qubit is lost; Bell-pair correlations survive.
- GHZ exhibits genuine 3-body nonlocality (Mermin inequality violations).

Question 1.3.3 – NUMERICAL (4 marks)**Post-Measurement State**

Bell state:

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

- (a) Write density matrix. (1 mark)
- (b) Compute purity $P = \text{Tr}(\rho^2)$. (1.5 marks)
- (c) If qubit 1 is measured as $|0\rangle$, find post-measurement state of qubit 2. (1.5 marks)

(a) Density matrix (1 mark)

$$\rho = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(b) Purity (1.5 marks)

Square the matrix:

$$\rho^2 = \frac{1}{2}\rho \quad \Rightarrow \quad \text{Tr}(\rho^2) = 1$$

→ pure state.

(c) Post-measurement state (1.5 marks)

Projector on qubit 1:

$$P_0 = |0\rangle\langle 0|$$

Applying measurement:

$$\rho_2 = |1\rangle\langle 1|$$

So qubit 2 collapses deterministically to $|1\rangle$.

Question 1.3.4 – THEORY (4 marks)

Design a quantum circuit generating

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

from $|00\rangle$. Show each step and justify the gates used.

Solution:

1. Apply Hadamard on qubit 1

$$|00\rangle \xrightarrow{H_1} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

2. Apply CNOT (1→2)

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \rightarrow \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle$$

3. Apply X on qubit 1

$$|\Phi^+\rangle \rightarrow \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

4. Apply Z on qubit 2

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \rightarrow \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) = |\Phi^-\rangle$$

Circuit:

$$|\Phi^-\rangle = (Z_2)(X_1)(\text{CNOT}_{1\rightarrow 2})(H_1)|00\rangle$$

Justification: H creates superposition; CNOT entangles; X and Z adjust bit/phase to reach the target Bell state.

Question 1.3.5 – SCENARIO NUMERICAL (4 marks)

Atmospheric teleportation of

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$$

uses an SPDC Bell pair with 10% depolarizing noise.

(a) Outline teleportation protocol and required classical bits. (1 mark)

(b) Bob receives state

$$\rho = 0.9|\psi\rangle\langle\psi| + 0.05I_2.$$

Compute fidelity. (2 marks)

(c) Bob's detector has 5% misidentification. Give final fidelity estimate. (1 mark)

(a) Protocol (1 mark)

Alice performs:

- Bell-measurement on (input qubit + her SPDC qubit),
- obtains 2 classical bits,
- sends them to Bob,
- Bob applies I , X , Z , or XZ accordingly.

(b) Fidelity (2 marks)

$$\rho_{\text{recv}} = 0.9|\psi\rangle\langle\psi| + 0.05I_2$$

$$F = \langle\psi|\rho|\psi\rangle = 0.9 + 0.05 = 0.95$$

Fidelity = 95%.

(c) Detector error (1 mark)

Correct detection probability = 95% Thus:

$$F_{\text{final}} = 0.95 \times 0.95 = 0.9025$$

Final fidelity 90%.

UNIT 4: QUANTUM COMPUTING

1-Mark Multiple Choice Questions (48 Questions)

Question 2.1.1 A classical bit can be in:

- A Only one definite state (0 or 1)
- B Superposition
- C Entangled state
- D A mixed quantum state

Answer: A

Classical bits are always definite 0 or 1.

Question 2.1.2 A qubit can exist in:

- A Only classical states
- B Superposition of $|0\rangle$ and $|1\rangle$
- C Only $|0\rangle$
- D Neither 0 nor 1

Answer: B

Quantum bits allow superposition.

Question 2.1.3 Which class represents quantum polynomial-time problems?

- A NP
- B BQP
- C P
- D EXP

Answer: B

BQP = Bounded-error Quantum Polynomial time.

Question 2.1.4 If $P = NP$, then:

- A All NP problems become easy
- B Classical computers become quantum
- C Quantum computing is impossible
- D $NP \not\subseteq P$ always

Answer: A

$P = NP$ implies polynomial-time solutions for NP problems.

Question 2.1.5 Deutsch–Jozsa determines if a function is:

- A Prime
- B Constant or Balanced
- C Even or Odd
- D Continuous or Discrete

Answer: B

Deutsch–Jozsa gives exponential speedup.

Question 2.1.6 Bernstein–Vazirani finds:

- A Hidden period
- B Hidden string
- C Modular inverse
- D Grover oracle

Answer: B

BV finds secret bitstring in one query.

Question 2.1.7 Simon's algorithm provides:

- A Classical speedup
- B Exponential quantum speedup
- C Logarithmic improvement
- D No speedup

Answer: B

Simon's algorithm gives exponential separation from classical.

Question 2.1.8 Simon's problem uses:

- A Hidden subgroup over $\mathbb{Z}_2^n$
- B QFT over reals
- C Grover iteration
- D Modular arithmetic

Answer: A

Simon's algorithm is a hidden-subgroup problem.

Question 2.1.9 Grover's algorithm gives:

- A Exponential speedup
- B Quadratic speedup
- C No improvement
- D Classical speed

Answer: B

Grover reduces search from $O(N)$ to $O(\sqrt{N})$.

Question 2.1.10 Grover iteration uses:

- A Hadamard only
- B Oracle + Diffusion
- C Fourier transform
- D SWAP gates

Answer: B

Grover = Phase Oracle + Inversion about mean.

Question 2.1.11 Grover–Radhakrishnan algorithm is:

- A Exact version of Grover
- B Partial Grover iterations
- C For factoring
- D For error-correction

Answer: B

Partial search: fewer iterations, probabilistic success.

Question 2.1.12 QPE estimates:

- A Eigenvalues of Hermitian operators
- B Classical functions
- C Memory usage
- D Qiskit latency

Answer: A

QPE finds phase corresponding to eigenvalue.

Question 2.1.13 QPE uses:

- A Controlled- U^{2^k}
- B SWAP gates
- C Bit flips only
- D Quantum RAM

Answer: A

Exponentially growing controlled unitaries required.

Question 2.1.14 QFT transforms computational basis states into:

- A Classical domain
- B Fourier frequency domain
- C Time domain
- D Entangled-only states

Answer: B

QFT = DFT applied to quantum amplitudes.

Question 2.1.15 Complexity of QFT is:

- A $O(2^n)$
- B $O(n^2)$
- C $O(n)$
- D $O(n!)$

Answer: B

QFT requires $n(n-1)/2$ controlled rotations.

Question 2.1.16 The key step in Shor's algorithm is:

- A Period finding
- B Classical search
- C Teleportation
- D Quantum measurement only

Answer: A

Order/period detection enables factoring.

Question 2.1.17 Shor's algorithm gives:

- A Quadratic speedup
- B Exponential speedup
- C No speedup
- D Slower performance

Answer: B

Factoring becomes polynomial-time.

Question 2.1.18 A Qiskit circuit begins with:

- A Classical operation
- B QuantumRegister creation
- C SWAP only
- D FFT

Answer: B

Qiskit = QuantumRegister + ClassicalRegister + QuantumCircuit.

Question 2.1.19 In Qiskit, measurement collapses:

- A Classical registers
- B Quantum state
- C Noise model
- D Backend info

Answer: B

Measurement yields classical results.

Question 2.1.20 Entanglement is created in Qiskit using:

- A X then Z
- B H then CNOT
- C SWAP only
- D Reset

Answer: B

Standard Bell state circuit sequence.

2-Mark Multiple Choice Questions (12 Questions for Unit 4)

Question 2.2.1 In Grover's search algorithm, a database contains $N = 8$ items (represented using 3 qubits), with exactly one marked item. Using the optimal iteration formula for Grover's algorithm, the number of iterations required to maximize the probability of measuring the marked state is:

- A 1
- B 2
- C 3
- D 4

Answer: B

SOLUTION (2 marks): (a) Grover optimal iteration count:

$$r = \left\lfloor \frac{\pi}{4} \sqrt{\frac{N}{m}} \right\rfloor$$

For $N = 8$, $m = 1$:

$$r = \left\lfloor \frac{\pi}{4} \sqrt{8} \right\rfloor = \lfloor 2.22 \rfloor = 2$$

(b) Success probability:

$$P = \sin^2((2r + 1)\theta) \approx 94\%$$

Thus, **2 iterations** give maximum success.

Question 2.2.2 For an $n = 5$ qubit input, the Deutsch–Jozsa algorithm distinguishes between constant and balanced functions using:

- A A single deterministic oracle call
- B Approximately 2^{n-1} classical queries
- C Two guaranteed oracle calls
- D A probabilistic decision process

Answer: A

SOLUTION (2 marks): (a) Quantum parallelism evaluates all 2^n inputs simultaneously. (b) A single oracle call followed by interference ensures that constant functions always yield $|0\rangle^{\otimes n}$, while balanced functions never do. Thus, 1 oracle call is enough.

Question 2.2.3 The Bernstein–Vazirani algorithm determines an n -bit hidden string a by using:

- A n oracle evaluations (one per bit)
- B A single quantum oracle evaluation
- C Exponentially many classical queries
- D A quantum phase estimation subroutine

Answer: B

SOLUTION (2 marks): (a) The oracle encodes $a \cdot x$ as a phase for all 2^n inputs simultaneously. (b) Hadamard gates convert this phase information into the computational basis, revealing a in one measurement. Thus, only **one** oracle call is needed.

Question 2.2.4 In Simon's algorithm, the measurement outcome y obtained after applying Hadamard gates satisfies:

A $y = r$

B $y \cdot r = 0 \pmod{2}$

C $y \oplus r = 1$

D $y = 0$

Answer: B

SOLUTION (2 marks): (a) Interference collapses the state into a subspace orthogonal to the hidden period r . (b) Each measurement gives a linear constraint $y \cdot r = 0$. Collecting $n - 1$ independent equations reveals r exponentially faster than classical.

Question 2.2.5 Quantum Phase Estimation (QPE) using $n_c = 4$ counting qubits provides a phase precision of:

A $1/2$

B $1/4$

C $1/16$

D $1/n_c$

Answer: C

SOLUTION (2 marks): (a) QPE resolves phases in steps of:

$$\Delta\phi = \frac{1}{2^{n_c}}$$

(b) For $n_c = 4$, precision is:

$$\Delta\phi = \frac{1}{16}$$

This allows extracting binary phase estimates accurately.

Question 2.2.6 Shor's algorithm relies on the Quantum Fourier Transform (QFT) mainly to:

- A Create entanglement
- B Convert periodicity into measurable frequency peaks
- C Perform quantum search
- D Compute classical gcd values

Answer: B

SOLUTION (2 marks): (a) Modular exponentiation creates a periodic amplitude pattern. (b) QFT transforms this pattern into frequency spikes located at integer multiples of $1/r$. This allows extracting the order r efficiently.

Question 2.2.7 A primary distinction between classical and quantum computation is that quantum computation:

- A Allows perfect copying of unknown states
- B Uses superposition and interference to process information
- C Always runs faster
- D Requires no measurements

Answer: B

SOLUTION (2 marks): (a) Superposition enables parallel evaluation of many inputs simultaneously. (b) Interference reinforces correct solutions (Grover) and suppresses incorrect ones. These features enable quantum speedup.

Question 2.2.8 The complexity class BQP represents problems:

- A Solvable deterministically in polynomial time
- B Efficiently solvable by quantum computers with bounded error
- C That are NP-complete
- D Requiring exponential quantum resources

Answer: B

SOLUTION (2 marks): (a) BQP = Bounded-error Quantum Polynomial time. (b) Includes problems such as Shor, Grover, Simon, and QPE-based tasks. These are solvable efficiently with error $\leq 1/3$.

Question 2.2.9 The Grover–Radhakrishnan (partial Grover) approach improves run-time by:

- A Removing the oracle stage
- B Reducing the number of Grover iterations at the cost of success probability
- C Using QFT instead of the diffusion operator
- D Ensuring amplitude becomes exactly 1

Answer: B

SOLUTION (2 marks): (a) Early termination reduces time significantly per run. (b) Multiple fast attempts often outperform a single long optimal Grover iteration in noisy hardware. Thus, a speed–accuracy tradeoff is used.

Question 2.2.10 The Quantum Fourier Transform (QFT) on n qubits consists mainly of:

- A Only Hadamard gates
- B Hadamard gates and controlled phase rotations
- C Pauli-X gates only
- D Diffusion operators

Answer: B

SOLUTION (2 marks): (a) Each qubit receives a Hadamard, followed by controlled- R_k rotations for phase encoding. (b) Complexity is $O(n^2)$, with optional SWAPs for bit reversal.

Question 2.2.11 A Bell state can be created in Qiskit using the sequence:

- A X then Z
- B H then CNOT
- C Z then H
- D SWAP then X

Answer: B

SOLUTION (2 marks): (a) The Hadamard initializes superposition: $(|0\rangle + |1\rangle)/\sqrt{2}$. (b) CNOT entangles the superposition with the second qubit, forming a Bell pair.

Question 2.2.12 In Qiskit, the Quantum Fourier Transform can be implemented directly using:

- A Classical FFT algorithms
- B `qiskit.circuit.library.QFT`
- C Measurement operations only
- D Reset and X gates

Answer: B

SOLUTION (2 marks): (a) Qiskit provides a built-in QFT circuit containing Hadamards, controlled- R_k gates, and SWAPs. (b) This allows integration into Shor, QPE, and other Fourier-based algorithms with minimal coding.

4-Mark Subjective Questions ($10 \times 4M = 40$ marks for Unit 4)

[Structure: 3 numerical per 40-mark block, 1 theory]

Question 2.3.1 - NUMERICAL (4 marks)

Grover's Algorithm Optimization

10-qubit database: $N = 2^{10} = 1024$ items with $m = 4$ marked items.

- (a) Calculate required iterations r (1.5 marks)
- (b) Find theoretical success probability P_s (1 mark)
- (c) Estimate fidelity loss with 0.1% oracle error per oracle call (1.5 marks)

DETAILED SOLUTION (4 marks total)**(a) Iterations (1.5 marks):**

The Grover rotation angle θ is given by

$$\sin \theta = \sqrt{\frac{m}{N}} = \sqrt{\frac{4}{1024}} = \frac{1}{16} = 0.0625.$$

Hence

$$\theta = \arcsin\left(\frac{1}{16}\right) \approx 0.06252 \text{ rad.}$$

The (continuous) optimal number of iterations is

$$r_{\text{opt}} = \frac{\pi}{4} \sqrt{\frac{N}{m}} = \frac{\pi}{4} \sqrt{\frac{1024}{4}} = \frac{\pi}{4} \cdot 16 = 4\pi \approx 12.566.$$

In practice choose an integer r near 12.57; $r = 12$ is commonly used (one may pick $r = 13$ if rounding up is desired).

(Full credit: show formula and compute ≈ 12.57 .)

(b) Success probability (1 mark):

The success probability after r Grover iterations is

$$P(r) = \sin^2((2r+1)\theta).$$

For $r = 12$:

$$(2r+1)\theta = 25\theta \approx 25 \times 0.06252 \approx 1.5635 \text{ rad,}$$

so

$$P(12) = \sin^2(1.5635) \approx 0.99995 \text{ (99.995\%).}$$

For $r = 13$: $P(13) \approx 98.6\%$. Thus $r = 12$ is effectively optimal here.

(c) Oracle error impact (1.5 marks):

Assume each oracle call has independent error probability $\varepsilon = 0.001$ (0.1%). With r iterations the oracle is called r times per run (neglecting small constant factors); probability of no oracle error is $(1 - \varepsilon)^r$. Thus probability of at least one oracle error in the run:

$$P_{\text{err}} = 1 - (1 - \varepsilon)^r.$$

For $r = 12$:

$$P_{\text{err}} = 1 - (1 - 0.001)^{12} \approx 1 - 0.98806 \approx 0.01194 \text{ (1.194\%).}$$

Approximate fidelity (successful outcome probability accounting for oracle errors):

$$F \approx P(12) \times (1 - P_{\text{err}}) \approx 0.99995 \times 0.98806 \approx 0.9880 \text{ (98.80\%).}$$

Comment: this multiplicative approximation is reasonable for small ε ; more detailed noise models would propagate phase errors through amplitude amplification and slightly change the number.

Marking: (1.5) correct r derivation; (1) correct $P(r)$ computation; (1.5) correct error propagation and final fidelity.

Question 2.3.2 - NUMERICAL (4 marks)**Shor's Algorithm: Period-Finding**

Factor $N = 21 = 3 \times 7$ using Shor's algorithm with $a = 2$.

- (a) Find the period r of $f(x) = 2^x \bmod 21$ (1.5 marks)
- (b) Extract factors from period using $\gcd(2^{r/2} \pm 1, 21)$ (1.5 marks)
- (c) Verify result (1 mark)

DETAILED SOLUTION (4 marks total)**(a) Period finding (1.5 marks):**

Compute powers of 2 modulo 21:

x	$2^x \bmod 21$
0	1
1	2
2	4
3	8
4	16
5	$32 \bmod 21 = 11$
6	$64 \bmod 21 = 1$

The sequence repeats at $x = 6$, so the period $r = 6$.

(b) Factor extraction (1.5 marks):

Since r is even, compute

$$2^{r/2} = 2^3 = 8.$$

Compute the gcds:

$$\gcd(2^{r/2} - 1, 21) = \gcd(7, 21) = 7, \quad \gcd(2^{r/2} + 1, 21) = \gcd(9, 21) = 3.$$

Thus the nontrivial factors are 3 and 7.

(c) Verification (1 mark):

$3 \times 7 = 21$, so factorization is correct.

Notes: In a quantum implementation period r would be obtained via QPE + QFT on the modular exponentiation unitary; this question performs the classical verification step.

Question 2.3.3 - NUMERICAL (4 marks)**Quantum Phase Estimation**

Estimate eigenphase of unitary $U = e^{i\sigma_z\pi/4}$ using a QPE circuit with $n = 3$ counting qubits initialized to $|0\rangle^{\otimes 3}$ and the system qubit prepared in eigenstate $|0\rangle$.

- (a) Design the QPE circuit; describe each component (1.5 marks)
- (b) Calculate phase precision with 3 counting qubits (1 mark)
- (c) What is the measured eigenvalue (phase) with uncertainty? (1.5 marks)

DETAILED SOLUTION (4 marks total)**(a) QPE circuit (1.5 marks):**

Circuit components and purpose:

1. **Counting register:** $n = 3$ qubits prepared in $|0\rangle^{\otimes 3}$; apply $H^{\otimes 3}$ to create equal superposition $\frac{1}{\sqrt{8}} \sum_{k=0}^7 |k\rangle$.
2. **Controlled powers:** Apply controlled- U^{2^j} from counting qubit j to the system qubit for $j = 0, 1, 2$ (least significant to most). Since U has eigenvalue $e^{i\phi}$ on $|0\rangle$, these operations imprint phases $e^{i2^j\phi}$ on the counting register.
3. **Inverse QFT:** Apply $\text{QFT}^\dagger$ on the counting register to convert phase information into computational-basis amplitudes.
4. **Measurement:** Measure the 3 counting qubits to obtain an n -bit estimate of the phase fraction $k/2^n$.

(b) Phase precision (1 mark):

With n counting qubits the discretization step (resolution) in radians is

$$\Delta\phi_{\text{step}} = \frac{2\pi}{2^n} = \frac{2\pi}{8} = \frac{\pi}{4}.$$

The typical guaranteed error (half-bin) is

$$\Delta\phi_{\text{error}} \leq \frac{\Delta\phi_{\text{step}}}{2} = \frac{\pi}{8}.$$

(c) Measured eigenvalue with uncertainty (1.5 marks):

For $U = e^{i\sigma_z\pi/4}$, the eigenstate $|0\rangle$ has eigenvalue $e^{+i\pi/4}$. The normalized phase (as a fraction of 2π) is

$$\varphi = \frac{\pi/4}{2\pi} = \frac{1}{8}.$$

In binary with $n = 3$ bits this exactly equals 0.001_2 (i.e. integer 1). So QPE will (ideally) output the bitstring $|001\rangle$, corresponding to phase $2\pi \times (1/8) = \pi/4$. The uncertainty (half-bin) in radians is $\leq \pi/8$; in fractional units of 2π that's $\leq 1/16$ (i.e. ± 0.0625 of a full turn). Thus the measured phase is

$$\varphi = \frac{1}{8} \pm \frac{1}{16} \quad (\text{in units of } 2\pi), \quad \text{or} \quad \pi/4 \pm \pi/8 \text{ (radians)}.$$

Remark: Because φ is exactly representable with 3 bits, QPE returns the exact value (neglecting noise).

Question 2.3.4 - THEORY (4 marks)**Grover's Diffusion Operator**

Derive the Grover diffusion operator $D = 2|s\rangle\langle s| - I$ and explain why repeated application amplifies the marked state amplitude. Use geometric interpretation in amplitude space.

DETAILED SOLUTION (4 marks total)**Derivation (1.5 marks):**

Start with the equal superposition

$$|s\rangle = H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle.$$

The diffusion operator is defined as a reflection about $|s\rangle$:

$$D = 2|s\rangle\langle s| - I.$$

One can show equivalently

$$D = H^{\otimes n}(2|0\rangle\langle 0| - I)H^{\otimes n},$$

i.e. apply Hadamards, reflect about $|0\rangle$, then undo Hadamards.

Geometric picture and amplification (1.5 marks):

The state space relevant to Grover search is spanned by two orthogonal directions:

$|w\rangle$ (normalized marked-subspace vector) and $|w^\perp\rangle$ (normalized superposition of unmarked states)

Express initial state $|s\rangle$ as

$$|s\rangle = \sin \theta |w\rangle + \cos \theta |w^\perp\rangle,$$

with $\sin \theta = \sqrt{m/N}$. The oracle O performs a reflection that flips the phase of $|w\rangle$ (reflect about $|w^\perp\rangle$); the diffusion D reflects about $|s\rangle$. Two successive reflections (oracle then diffusion) equal a rotation by angle 2θ in the two-dimensional subspace spanned by $|w\rangle, |w^\perp\rangle$. Thus each Grover iteration rotates the state closer to $|w\rangle$, increasing the marked amplitude. After r iterations the marked amplitude becomes $\sin((2r+1)\theta)$, which approaches 1 when $(2r+1)\theta \approx \pi/2$.

Convergence choice of r (1 mark):

Solving $(2r+1)\theta \approx \pi/2$ gives $r \approx \frac{\pi}{4\theta} \approx \frac{\pi}{4}\sqrt{N/m}$, the usual Grover iteration count. Repeated application amplifies the marked amplitude until the rotation overshoots; hence picking r close to $\frac{\pi}{4}\sqrt{N/m}$ is optimal.

Marking: 1.5 marks for operator form and equivalent circuit; 1.5 marks for 2D geometric rotation argument; 1 mark for linking to $r \approx \frac{\pi}{4}\sqrt{N/m}$.

Question 2.3.5 - SCENARIO-NUMERICAL (4 marks)**Quantum Database Search: Decryption Key Recovery**

Story: A 16-bit key (search space $N = 2^{16} = 65,536$). Classical key verification takes $10\mu\text{s}$. Quantum resources: Grover search with 16 qubits, oracle call cost $10\mu\text{s}$, single/two-qubit gates 100 ns , decoherence limit 100 ms , time budget 1 s , required success probability 85% .

- (a) (1.5 marks) Calculate Grover iterations needed and theoretical success probability.
- (b) (1 mark) Estimate total runtime for the quantum approach. Will it fit within coherence/time budget?
- (c) (1.5 marks) If 64 wrong keys are pre-rejected, recalculate iterations and compute speedup vs classical brute-force.

DETAILED SOLUTION (4 marks total)**(a) Iterations and success (1.5 marks):**

With $N = 2^{16} = 65,536$ and single marked key $m = 1$,

$$r_{\text{opt}} = \frac{\pi}{4} \sqrt{\frac{N}{m}} = \frac{\pi}{4} \cdot 256 \approx 201.06.$$

Pick $r = 201$ iterations. The initial angle $\theta = \arcsin(\sqrt{1/N}) \approx 1/256$ (in radians approximately), so the theoretical success probability after $r = 201$ is

$$P(r) = \sin^2((2r+1)\theta) \approx \text{very close to } 1.$$

A simple approximation using $P \approx 1 - \frac{m}{N}$ is very close to 1 here; numerically $P \gtrsim 99.99\%$, well above the 85% requirement.

(b) Runtime feasibility (1 mark):

Major cost is oracle calls: each oracle = $10 \mu\text{s}$. For $r = 201$:

$$T_{\text{oracle}} = 201 \times 10 \mu\text{s} = 2010 \mu\text{s} = 2.01 \text{ ms}.$$

Add diffusion/gate overhead: even if diffusion per iteration costs $\sim 1 \mu\text{s}$ (conservative), total diffusion time $\approx 201 \mu\text{s}$. Add initialization + measurement (negligible). Total quantum runtime $\approx 2.3 \text{ ms}$. This is far below the decoherence limit (100 ms) and the 1 s time budget.

Conclusion: Quantum approach completes easily within coherence/time budgets.

(c) With 64 pre-rejected keys (1.5 marks):

Pre-filtered space $N' = 65,536 - 64 = 65,472$. New optimal iterations:

$$r' = \frac{\pi}{4} \sqrt{N'} \approx \frac{\pi}{4} \times \sqrt{65,472} \approx \frac{\pi}{4} \times 255.9 \approx 200.5 \approx 200.$$

So only ~ 1 iteration saved (very small change). Classical average brute-force time:

$$T_{\text{classical}} = \frac{N}{2} \times 10 \mu\text{s} = 32,768 \times 10 \mu\text{s} = 327,680 \mu\text{s} = 327.68 \text{ ms}.$$

Quantum time $\approx 2.3 \text{ ms} \Rightarrow$ speedup factor

$$\text{Speedup} \approx \frac{327.68 \text{ ms}}{2.3 \text{ ms}} \approx 142 \times (\text{order of } 100\text{--}150).$$

(Small differences in gate/oracle cost assumptions change the numeric factor slightly.) Even after pre-filtering, quantum Grover provides a very large speedup.

Marking: (1.5) correct r and P reasoning; (1) runtime estimates and feasibility; (1.5) recomputed r' , classical time and speedup.